Kambiz Ghoorchian

Professional Summary

Highly experienced Senior Data Science professional with over 10 years of experience in leading data-driven projects across various domains. Proven track record in developing and implementing advanced analytics solutions. Passionate about leveraging AI, cloud and big data technologies to drive strategic decision making and solve complex business challenges.

Work Experiences

2023 - now **Senior Data Scientist**, *Scandinavia Enskilda Banken (SEB)*, Stockholm, Sweden

2021 - 2024 **Industrial Research Co-Lead at Research Arena for Media and Language (WARA M&L)**, *in collaboration with Wallenberg AI, Autonomous Systems and Software Program (WASP)*, Stockholm, Sweden

2019 - 2023 **Data Scientist**, *Scandinavia Enskilda Banken (SEB)*, Stockholm, Sweden

Education

2014 - 2019 **Ph.D. Scholar**, *KTH Royal Institute of Technology*, Sweden
- Thesis: Scalable Stochastic Optimization using Distributional Semantics and Graph Analytics for Applications in Natural Language Processing.
- Adviser: Prof. Magnus Boman

2009 - 2011 **M.Sc. in Computational Linguistics**, *Katholic University of Leuven (KULeuven)*, Belgium
- Thesis: Co-reference/Anaphora Resolution in Scientific Text.
- Advisors: Prof. Frank Van Eynde, Dr. Ulrich Schaefer

2007 - 2009 **M.Sc. in Distributed Systems**, *KTH Royal Institute of Technology*, Sweden
- Thesis: Distributed Cross Document Co-reference Resolution.
- Advisers: Prof. Seif Haridi, Dr. Fatemeh Rahimian

Projects and Skills

Technical Projects

- Portfolio Optimization using RL-based PPO algorithm with gaussian reward and quadratic risk modeling.
- Rogue trading identification with unsupervised anomaly detection using isolation forest on FX-Trading data.
- End-to-end LLM based Q&A system with in-memory RAG engine for assisting HR staff on resolving internal employee's HR requests.
- Multivariate stochastic linear regression modeling for cashflow prediction to deliver short, mid, and long-term cashflow insights for enterprise costumers.

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- Identifying malicious or incorrect counter-party structure identification within complex networked of customer product contracts large-scale graph clustering.
- Fraud detection for credit card transactions using supervised and unsupervised machine learning approaches using random forest and graph clustering.

Regulations and Governance

- EU AI Act working group - Working as a senior machine learning and highly skilled technical developer expert together with risk, and legal teams to understand and establish proper processes to ensure organization's compliance with the new European AI regulations.

Academic and Research

PhD Student Supervision

- Ciwan Cylan - Towards Unsupervised, Analysable and Scalable Node Embedding Models for Transaction Networks (Dec 2025)

Masters Supervision

- Forsman, Tim. & Tomassini, Valentina - Portfolio Optimization with Reinforcement Learning (Exp 2026)
- Farrahi, Sheila. - Feature Construction of Multi-sets for Machine Learning Application (2024)
- Kjellberg, Gustav. - Federated Learning for Natural Language Processing (2022)
- Hammarstedt, Johan. - Synthetic Graph Generation at scale : A novel framework for generating large graphs using clustering, generative models and node embeddings(2020)

Skills, technologies and tools

- Active open source contributor (e.g. huggingface)
- Machine Learning: Scikit-learn, Pandas, Numpy, PyTorch, Tensorflow, Scipy
- NLP tools: nltk, openNLP, StanfordNLP, anaconda, SparkML
- Cloud Platforms: Google Cloud Platform (GCP)
- Programming Languages: Python, R, Scala, java, SQL
- Big Data Technologies: Spark, Flink, Hadoop, MapReduce, docker container.
- Embedding and generative models: word2vec, Bert, Gemini, GPT.
- Infrastructure as code: terraform
- code integration, testing, and deployment using CI/CD with gitlab and github on GCP

Certificates

- Fine Tune BERT for Text Classification with TensorFlow
- Convolutional Neural Networks in TensorFlow
- Natural Language Processing in TensorFlow
- Sequences, Time Series and Prediction
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning
- Sequence Models
- Convolutional Neural Networks

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- Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization
- Structuring Machine Learning Projects
- Neural Networks and Deep Learning

Publications

2024 C. Ceylan, K. Ghoorchian, D. Kragic, "Scalable Unsupervised Feature Selection with Reconstruction Error Guarantees via QMR Decomposition", CIKM '24: Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, Pages 3658 - 3662, https://doi.org/10.1145/3627673.3679994

2022 C. Ceylan, K. Ghoorchian, D. Kragic, "Digraphwave: Scalable Extraction of Structural Node Embeddings via Diffusion on Directed Graphs", arXiv:2207.10149

2018 K. Ghoorchian, M. Sahlgren, "GDTM: Graph-based Dynamic Topic Models", Submitted to The Web Conference (WWW'19), San Francisco, CA, USA.

2017 K. Ghoorchian, S. Girdzijauskas, F. Rahimian, "DeGPar: Large Scale Topic Detection using Node-Cut Partitioning on Dense Weighted Graphs", IEEE-ICDCS'17, Atlanta, GA, USA.

2015 K. Ghoorchian, F. Rahimian, S. Girdzijauskas, "Semi-Supervised Multiple Disambiguation", In Proceedings of IEEE-BigDataSE'15, Helsinki, Finland.

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